

In [5]:

```
#conditional statement (Decision making statement)
num = int(input("Enter a number"))
if num%2 == 0:
    print("Even number")
```

Enter a number42

Even number

In [6]:

```
a = int(input("Enter a number"))
b = int(input("Enter a number"))
c = int(input("Enter a number"))
if a>b and a>c:
    print("a is greater ", a)
if b>a and b>c:
    print("b is greater ", b)
if c>a and c>b:
    print("c is greater ", c)
```

Enter a number56

Enter a number45

Enter a number32

a is greater 56

In [9]:

```
age = int(input("Enter your age"))
if age< 18:
    print("Not eligible to vote")
else:
    print("Yes!! Eligible to vote")
```

Enter your age45

Yes!! Eligible to vote

In [17]:

```
marks = int(input("Enter your marks"))
if marks<100 and marks>90:
    print("Grade is A+")
elif marks<90 and marks>70:
    print("Grade is A")
elif marks<70 and marks>50:
    print("Grade is B+")
elif marks<50 and marks>0:
    print("Not elible to grade")
else:
    print("Enter marks in range of 0 to 100")
```

Enter your marks-34

Enter marks in range of 0 to 100

In [2]:

```
values = [15,25,33,24,215]
for value in values:
    print("Sanjay ", value)
```

Sanjay 15
Sanjay 25
Sanjay 33
Sanjay 24
Sanjay 215

In [3]:

```
for value in range(len(values)):
    print("hello")
```

hello
hello
hello
hello
hello

In [24]:

```
print(len(values))
```

5

In [4]:

```
print(values)
for value in range(len(values)):
    values.append(values[value]+2)
print(values)
```

[15, 25, 33, 24, 215]
[15, 25, 33, 24, 215, 17, 27, 35, 26, 217]

In [17]:

```
numbers = [1,2,3,4,5,6,7,8,9,10]
sum = 0
for num in range(len(numbers)):
    sum = sum + numbers[num] ** 2 # sum of the squares of the number
print("Sum of the list = " , sum)
```

Sum of the list = 385

In [19]:

```
sum = 0
for num in range(len(numbers)):
    if numbers[num]%2 ==0:
        sum = sum + numbers[num] ** 2 # sum of the squares of the even number
print("Sum of the list = " , sum)
```

Sum of the list = 220

In [20]:

```
for i in range(1,11):  
    print(i)
```

1
2
3
4
5
6
7
8
9
10

In [24]:

```
import random  
for i in range(1,10):  
    print(i+random.randint(20,30))
```

23
31
30
31
31
26
34
29
29

In [30]:

```
var = [1,2,3,4,5]  
for i in range (1,3):  
    print("Next iteration = " , i)  
    for num in range(len(var)):  
        print(var[num])
```

Next iteration = 1
1
2
3
4
5
Next iteration = 2
1
2
3
4
5

In [46]:

```
row = 6
for x in range(1,row + 1):
    for y in range(1,x + 1):
        print("*", end=' ')
    print('')
```

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

In [55]:

```
row = 6
for x in range(1,row + 1):
    for y in range(row,x-1,-1):
        print("*", end=' ')
    print('')
```

```
* * * * * *
* * * * *
* * * *
* * *
* *
*
*
```

In []:

```
names = ["Rohan", "Sohan", "Rajiv", "Sanjay", "Mohit"]
#names = []
i = 0
while i < len(names):
    print(names[i])
    i += 1
```

In [7]:

```
name = "Ayaan"
print(len(name))
j = 0
while j < len(name):
    if name[j] == 'a':
        j += 1
        continue
    print(name[j])
    j += 1
```

```
5
A
y
n
```

In [10]:

```

name = "Ayaan"
print(len(name))
j = 0
while j < len(name):
    j += 1
    pass
print("j = " , j)

```

5
j = 5

In []:

```

# infinite loop
#while(1):
#    print("hello")

```

In [11]:

```

row = 10
space = 2 * row - 2
for i in range(0, row):
    for j in range(0, space):
        print(end=" ")
    space = space - 2
    for j in range(0, i + 1):
        print("*", end=" ")
    print("")

```

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * * *
* * * * * * *
 * * * * * *
  * * * * *
   * * * *
    * * *
     * *
      *

```

In [18]:

```

rows = 6
k = 2 * rows - 2
# Outer Loop in reverse order
for i in range(rows, -1, -1):
    # Inner Loop will print the number of space.
    for j in range(k, 0, -1):
        print(end=" ")
    k = k + 1
    # This inner loop will print the number o stars
    for j in range(0, i + 1):
        print("*", end=" ")
    print("")

```

```

* * * * *
* * * * *
* * * *
* * *
* *
*

```

In [21]:

```

row = 10
space = 2 * row - 2
for i in range(0, row):
    for j in range(0, space):
        print(end=" ")
    space = space - 2
    for j in range(0, i + 1):
        print("*", end=" ")
    print("")

k = 2 * rows - 2
# Outer Loop in reverse order
for i in range(rows, -1, -1):
    # Inner Loop will print the number of space.
    for j in range(k, 0, -1):
        print(end=" ")
    k = k + 1
    # This inner loop will print the number o stars
    for j in range(0, i + 1):
        print("*", end=" ")
    print("")

```

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * * *
* * * * * * *
 * * * * *
  * * * *
   * * *
    * *
     *

```

In [26]:

```
rows = 6
for i in range(0, rows):
    # This inner loop will print the stars
    for j in range(0, i + 1):
        print(i+j, end=' ')
    # Change line after each iteration
    print(" ")

# For second pattern
for i in range(rows + 1, 0, -1):
    for j in range(0, i - 1):
        print(i+j, end=' ')
    print(" ")
```

```
0
1 2
2 3 4
3 4 5 6
4 5 6 7 8
5 6 7 8 9 10
7 8 9 10 11 12
6 7 8 9 10
5 6 7 8
4 5 6
3 4
2
```

In [24]:

```
row = 6
for x in range(1, row + 1):
    for y in range(1, x + 1):
        print("*", end=' ')
    print('')

for x in range(1, row + 1):
    for y in range(row, x-1, -1):
        print("*", end=' ')
    print('')
```

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
* * * * * *
* * * * *
* * * *
* * *
* *
*
```

In []: